AWS Cloud Server Setup

CS431 - Web Server Design

Spring 2026

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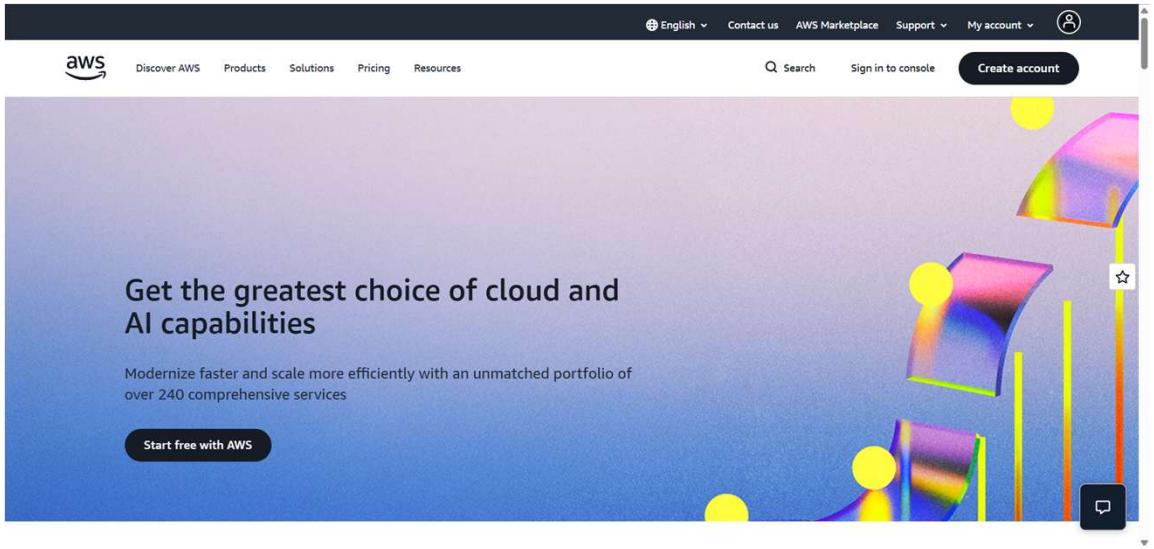
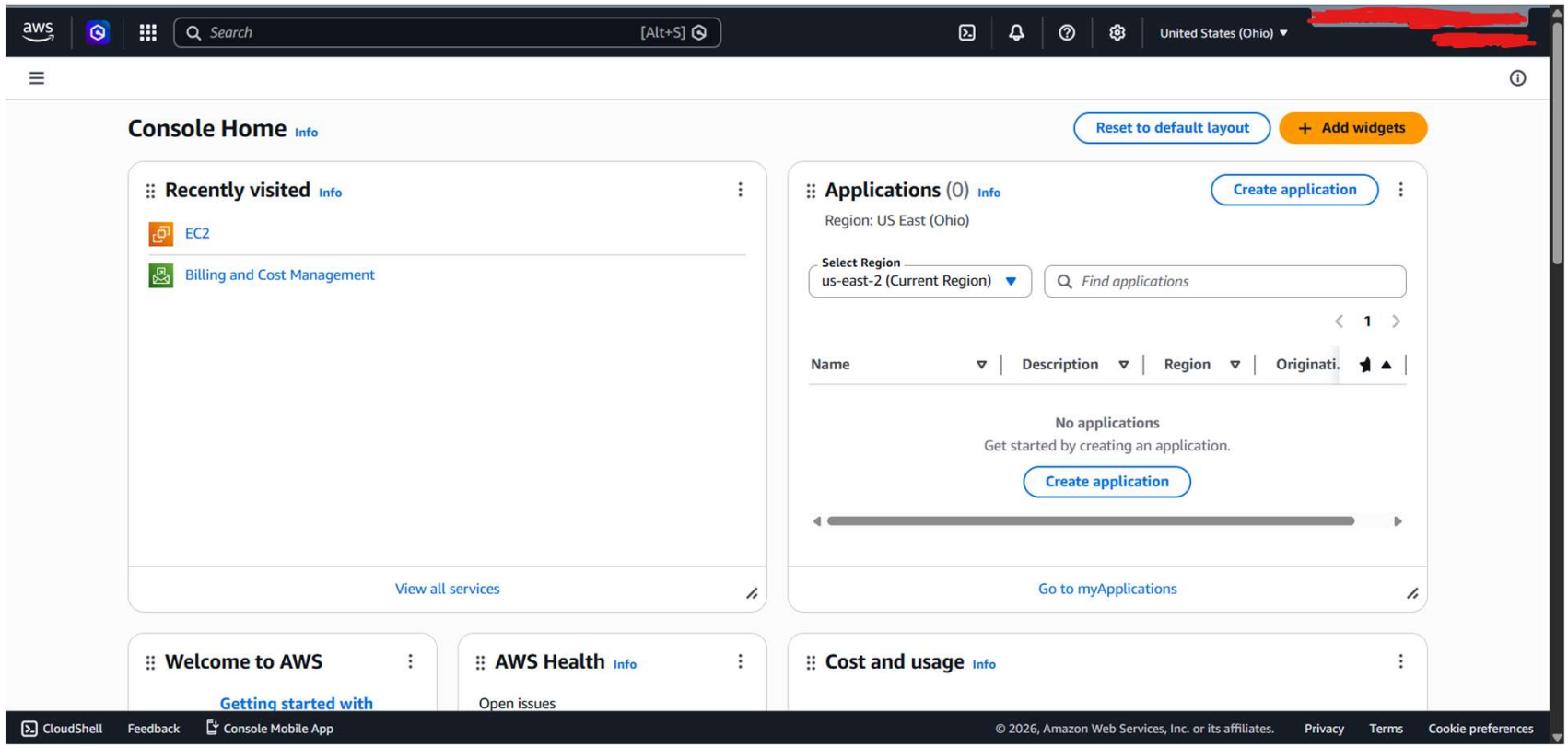
Creating an AWS Account

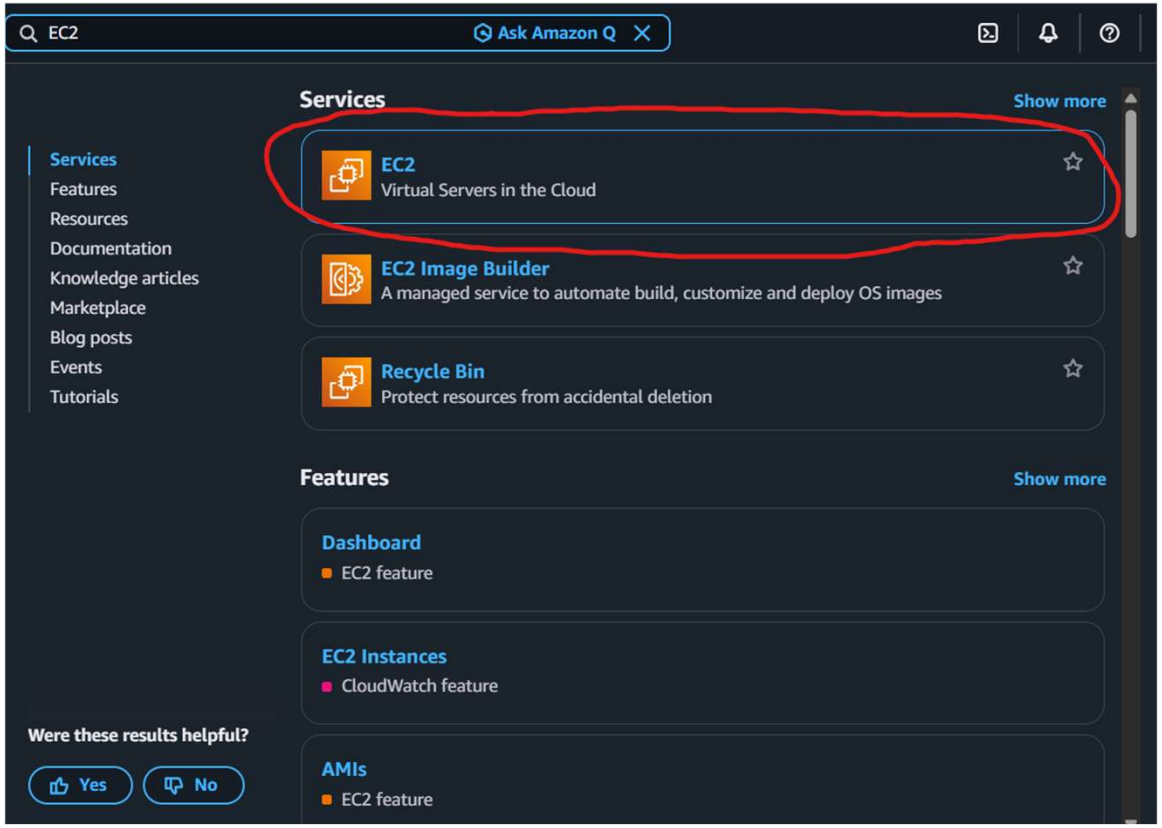
- To get started, make an AWS account at: https://aws.amazon.com/



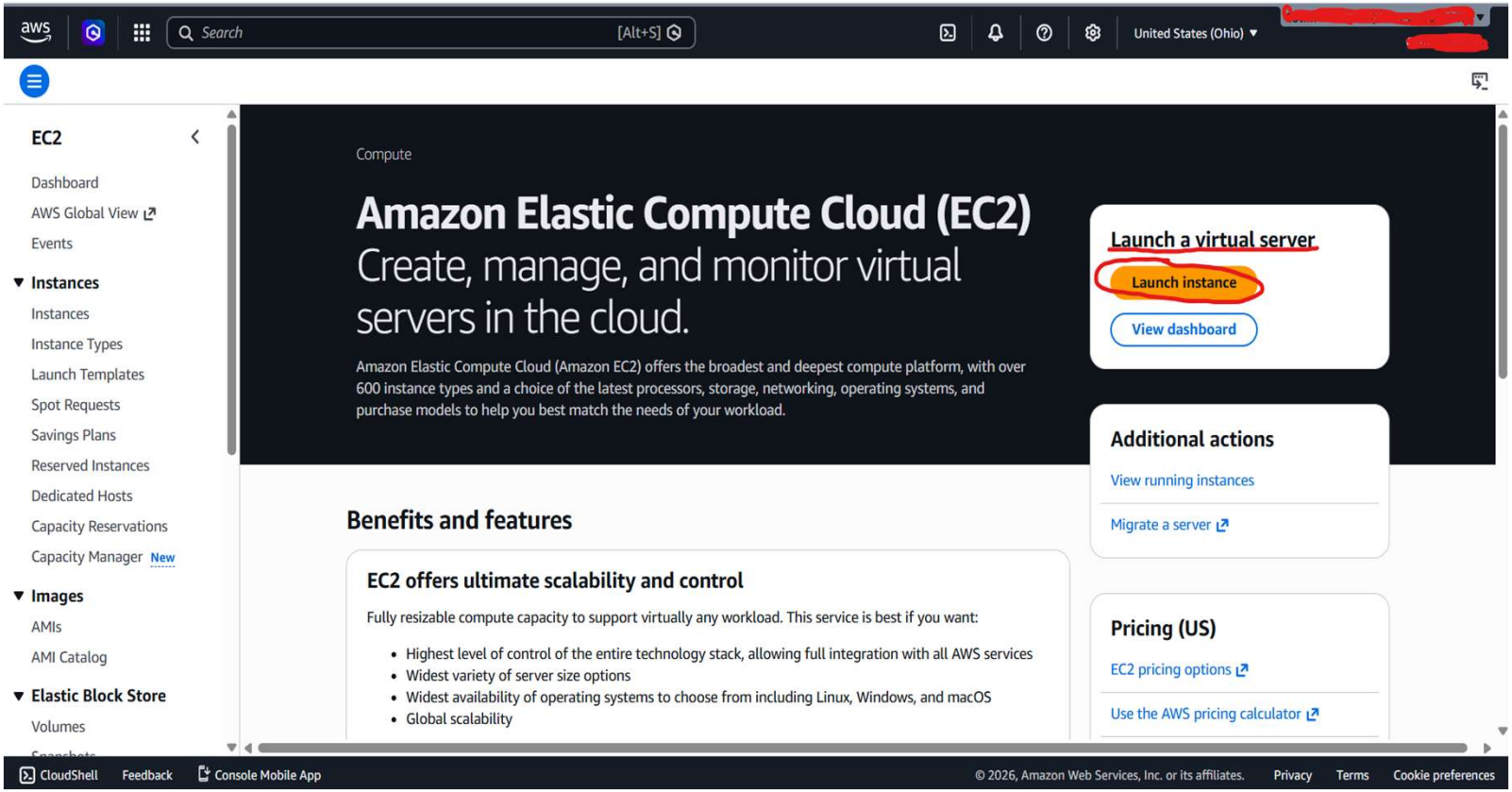
AWS Console Dashboard



Find EC2 in the Search-Bar

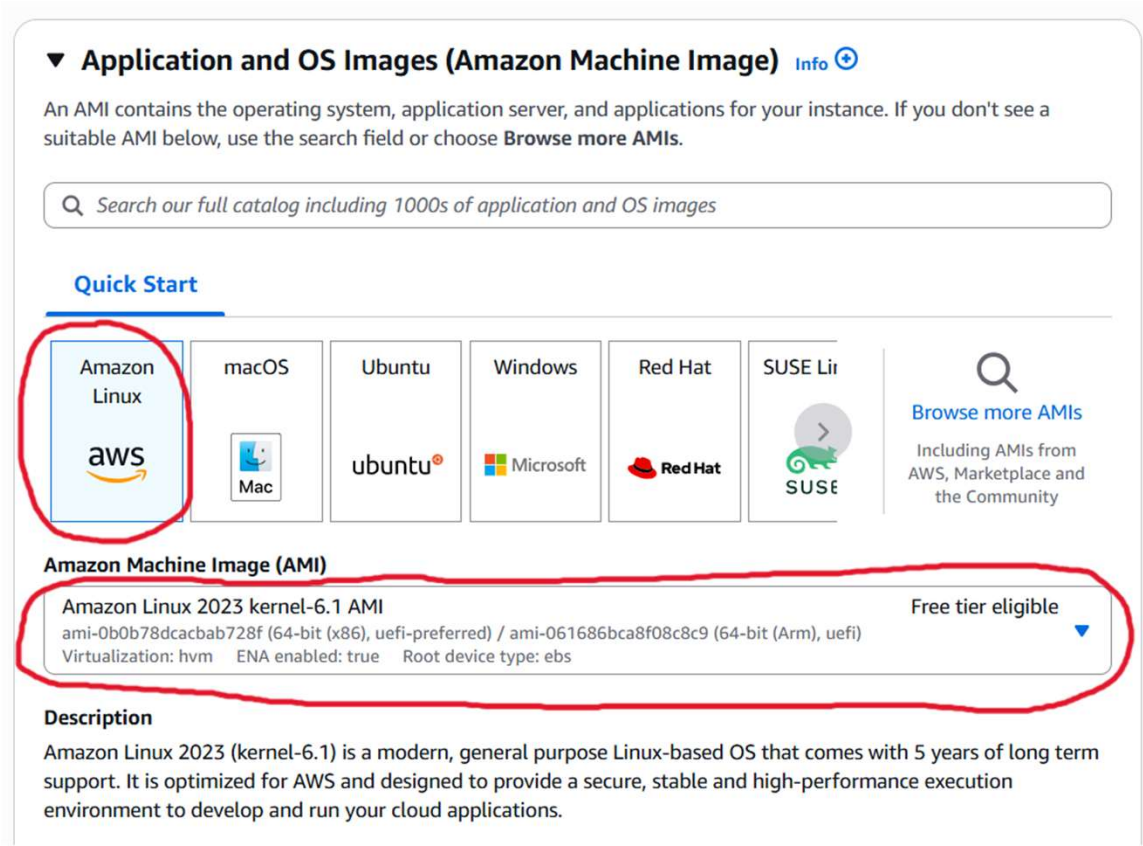


Create an EC2 Server Instance

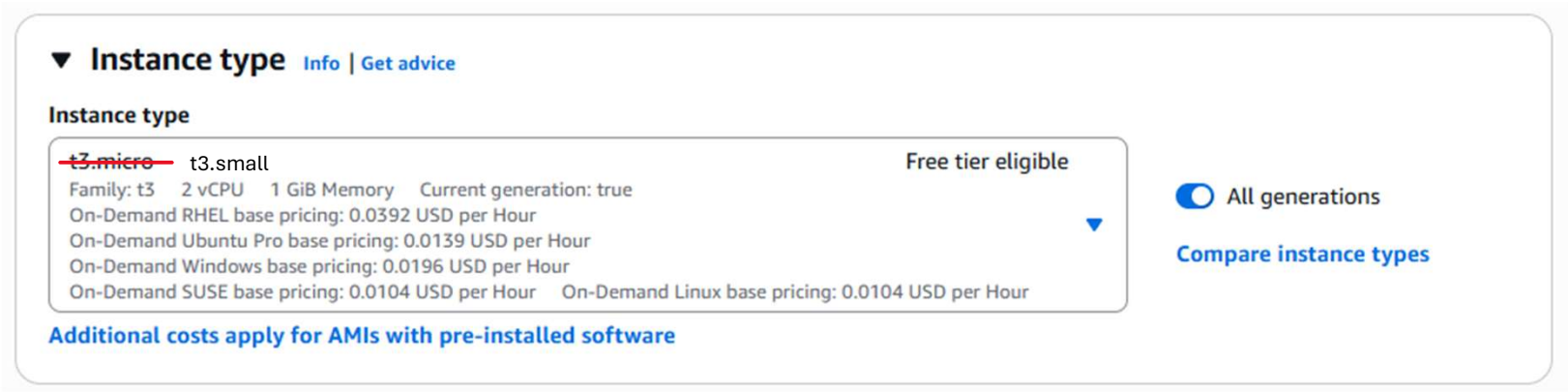


Step One: Name the Server

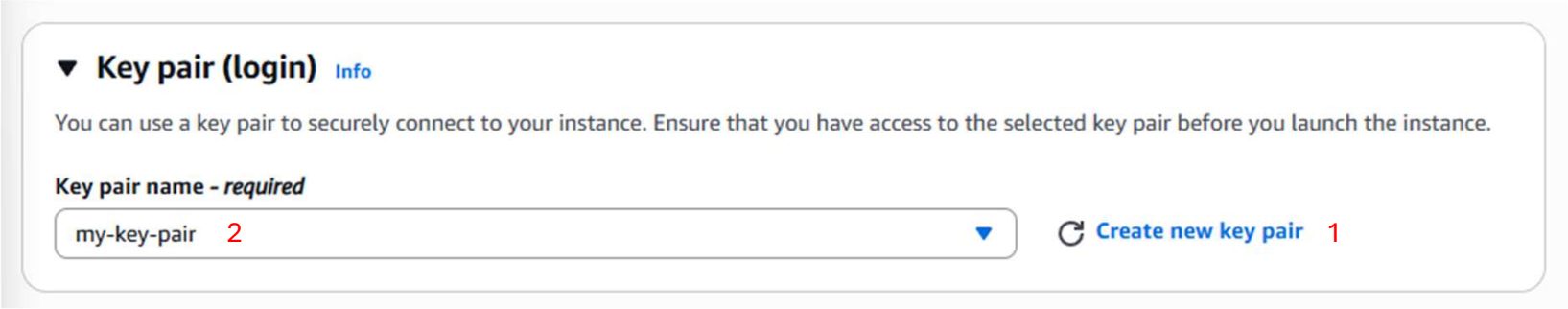
Step Two: Choose an OS & AMI



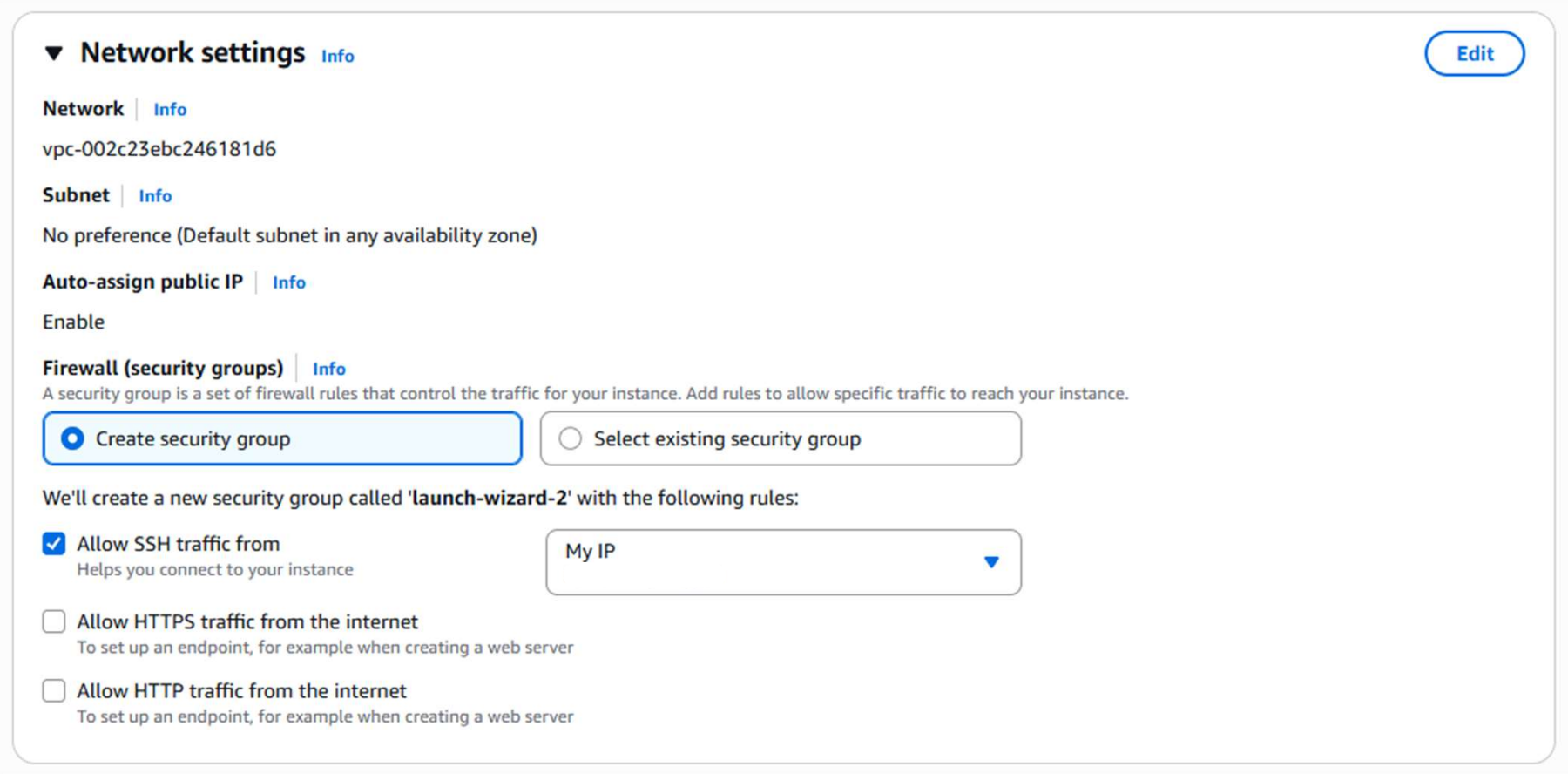
Step Three: Selecting an Instance Type



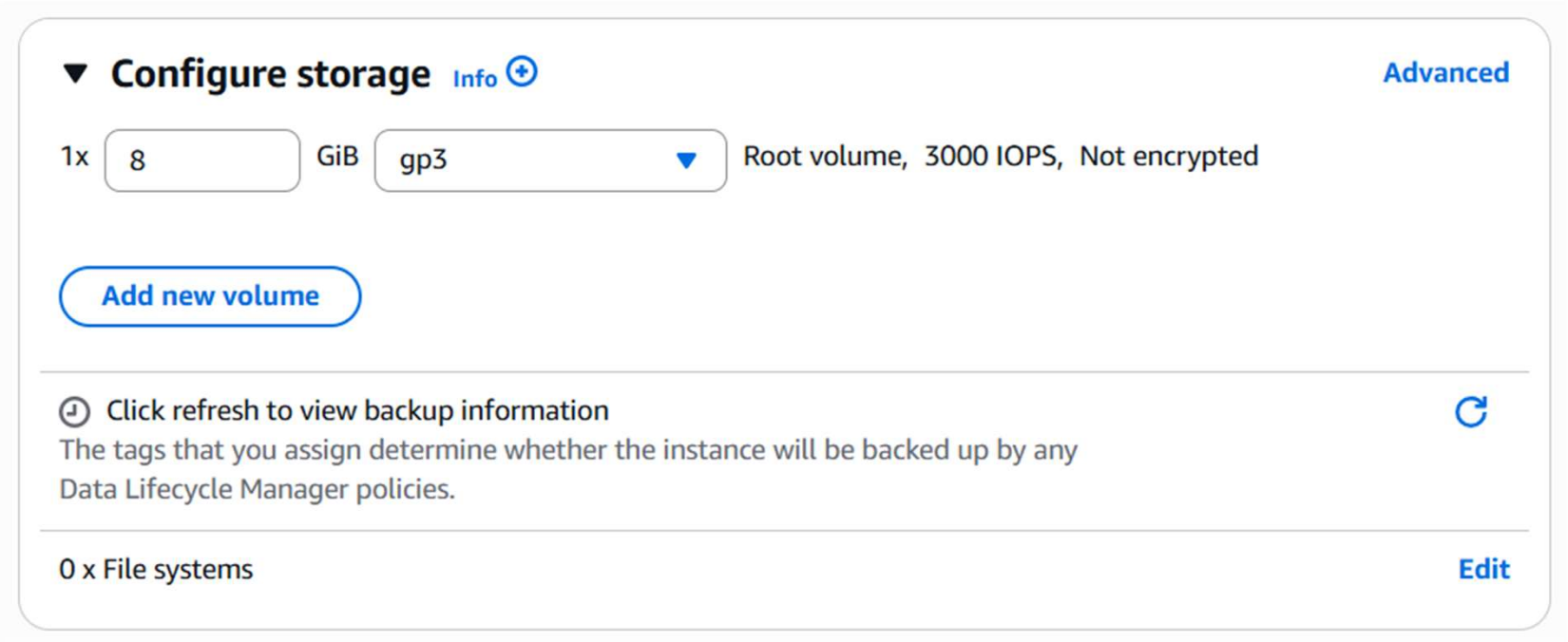
Step Four: Setup Login Key-Pair



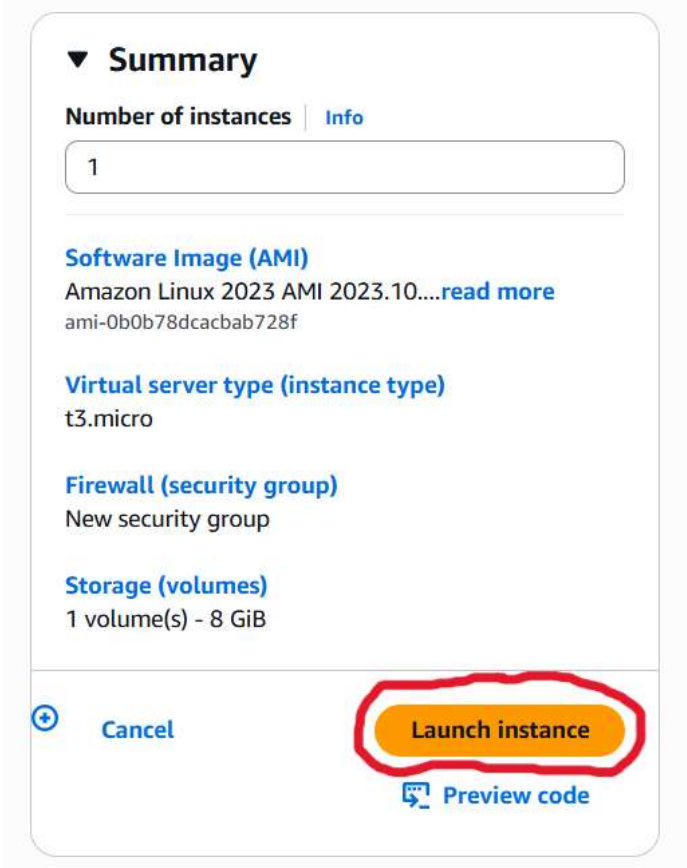
Step Five: Configure the Network



Step Six: Configure the Storage

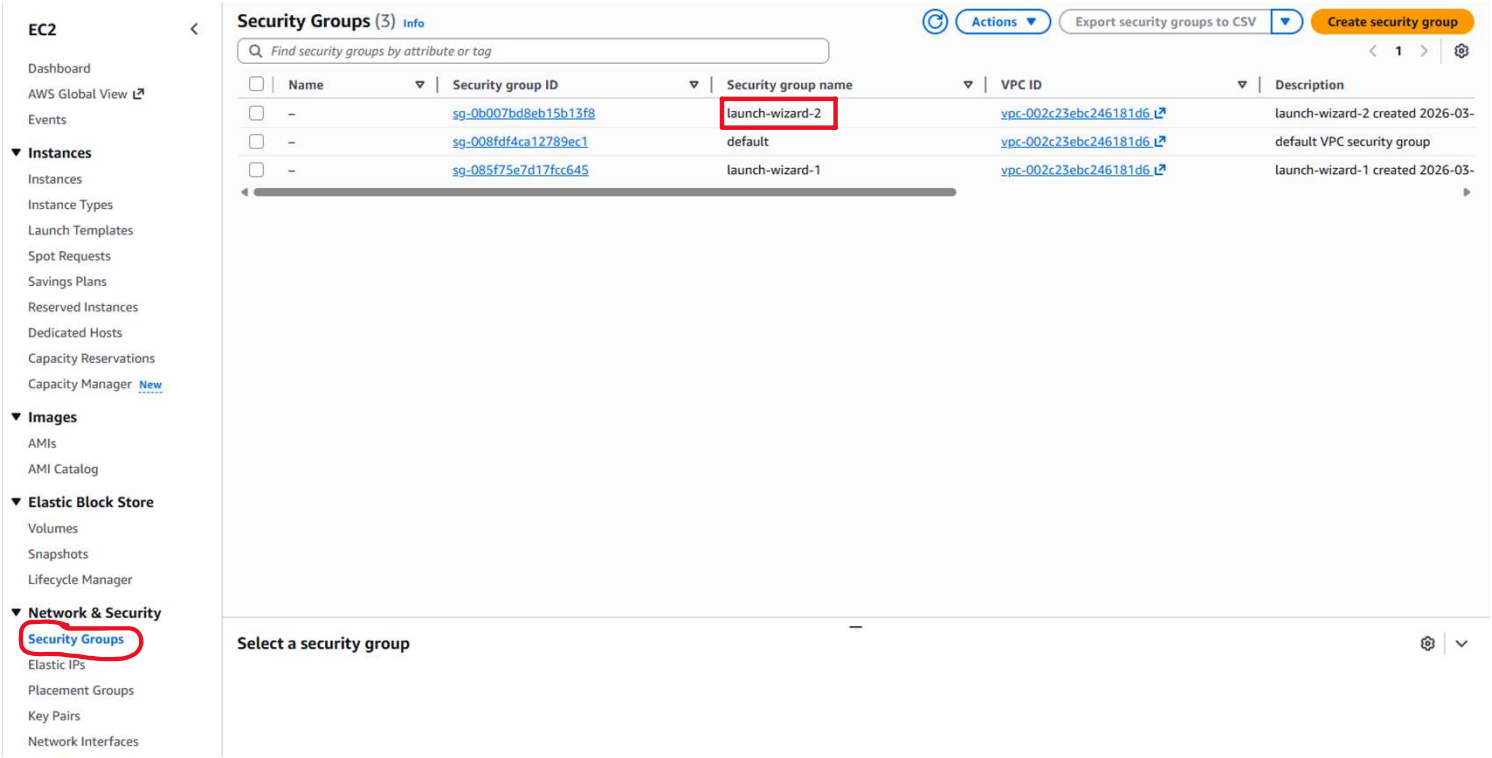


Step Seven: Launch the Instance



Step Eight: Assure the Launch was Successful

Step Nine: Edit the SG's Inbound Rules



Security Groups (1/3) Info

Actions | Export security groups to CSV | Create security group

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description
–	sg-0b007bd8eb15b13f8	launch-wizard-2	vpc-002c23ebc246181d6	launch-wizard-2 created 2026-03-
–	sg-008fdf4ca12789ec1	default	vpc-002c23ebc246181d6	default VPC security group
–	sg-085f75e7d17fcc645	launch-wizard-1	vpc-002c23ebc246181d6	launch-wizard-1 created 2026-03-

sg-0b007bd8eb15b13f8 - launch-wizard-2

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (1)

Manage tags | Edit inbound rules

Search

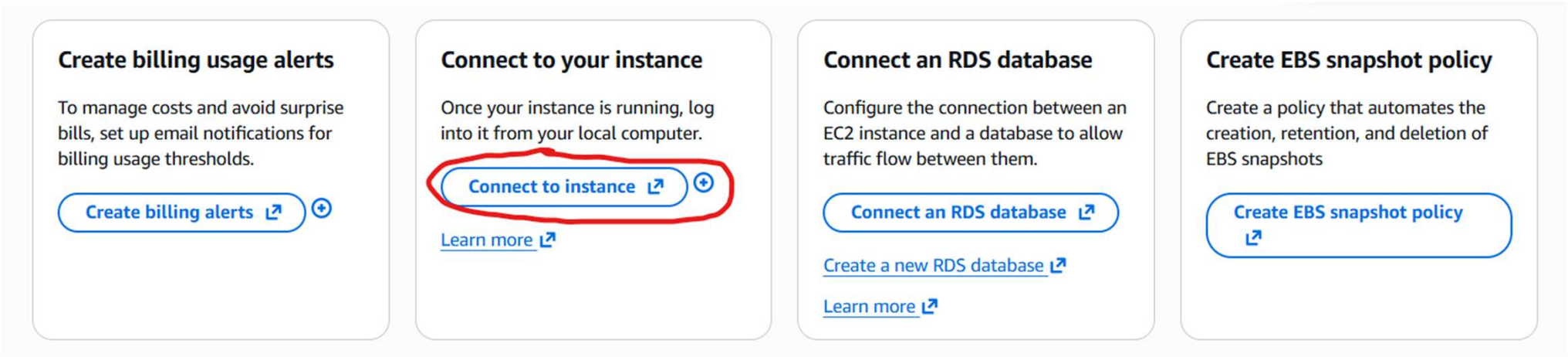
Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-0b07f220e5b935e61	IPv4	SSH	TCP	22	70.161.21

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-0e49ffd17ed14ccde	HTTP	TCP	80	Custom			Delete
sgr-0b07f220e5b935e61	SSH	TCP	22	Custom			Delete

Add rule

Step Ten: Connect to the Instance



EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

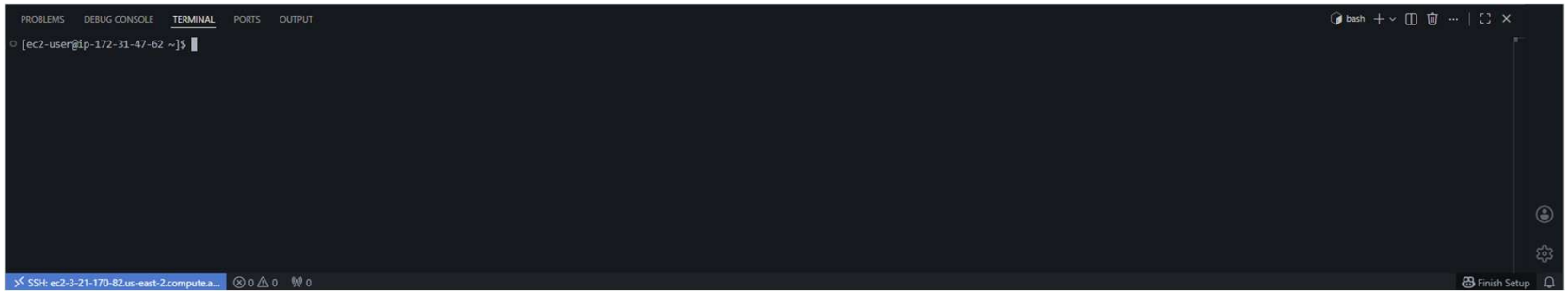
Instance ID
i-0a0af7142a7c5b5bf (CS431_A-02)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is my-key-pair.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
   chmod 400 "my-key-pair.pem"
4. Connect to your instance using its Public DNS:
   ec2-3-21-170-82.us-east-2.compute.amazonaws.com

Example:
ssh -i "my-key-pair.pem" ec2-user@ec2-3-21-170-82.us-east-2.compute.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

We’re In! *(used Remote SSH extension)*



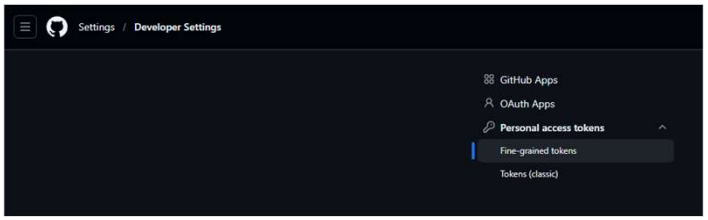
Step Eleven: Install Git & Docker

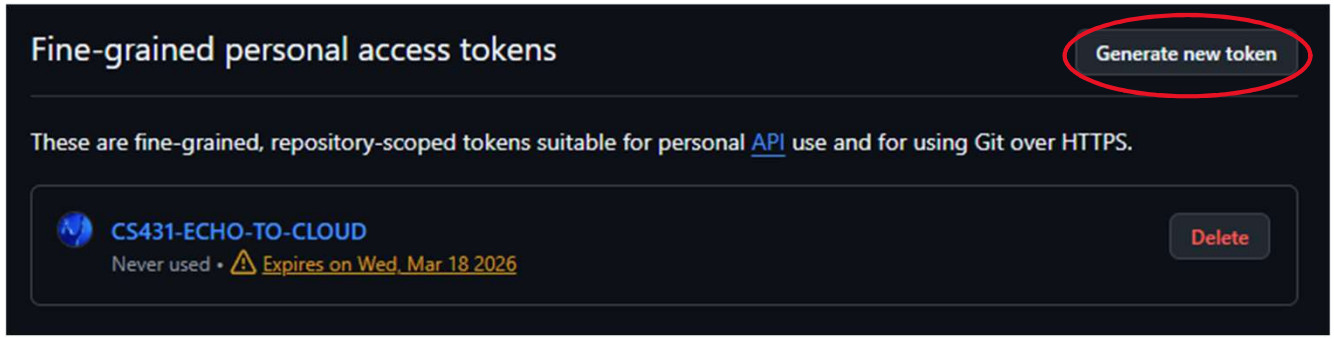
```
PROBLEMS    DEBUG CONSOLE    TERMINAL    PORTS    OUTPUT

[ec2-user@ip-172-31-47-62 ~]$ git --version
git version 2.50.1
[ec2-user@ip-172-31-47-62 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-47-62 ~]$
```

Step Twelve: Retrieve our Echo-Server

Twelve (a): Generate a PAT





Twelve(b): Clone the Repository to the Server

```
[ec2-user@ip-172-31-47-62 ~]$ git clone https://github.com/Colin-TM/ScratchServer.git
Cloning into 'ScratchServer'...
Username for 'https://github.com': Colin-TM
Password for 'https://Colin-TM@github.com': PAT Here
remote: Enumerating objects: 522, done.
remote: Counting objects: 100% (522/522), done.
remote: Compressing objects: 100% (282/282), done.
remote: Total 522 (delta 178), reused 444 (delta 100), pack-reused 0 (from 0)
Receiving objects: 100% (522/522), 431.95 KiB | 3.96 MiB/s, done.
Resolving deltas: 100% (178/178), done.
[ec2-user@ip-172-31-47-62 ~]$
```

Step Thirteen: Gather Docker Components

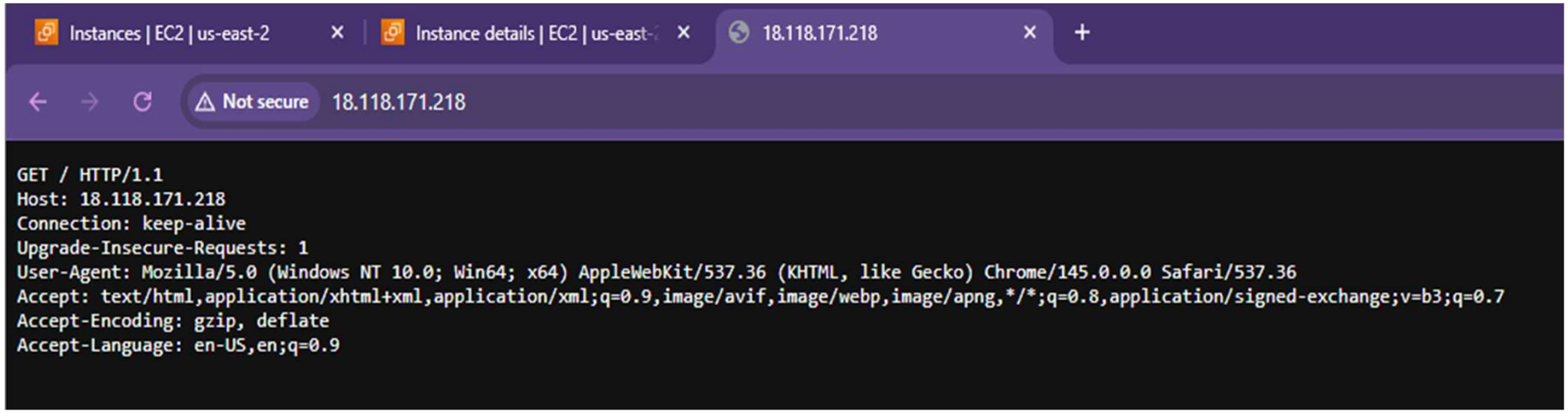
```
PROBLEMS    DEBUG CONSOLE    TERMINAL    PORTS    OUTPUT

[ec2-user@ip-172-31-47-62 ScratchServer]$ sudo docker ps -a
CONTAINER ID   IMAGE        COMMAND                  CREATED         STATUS                       PORTS     NAMES
ccb1102e8558   echo_image   "/__cacert_entrypoin…"   5 minutes ago   Exited (130) 13 seconds ago            echo_container
[ec2-user@ip-172-31-47-62 ScratchServer]$ sudo docker image ls -a
REPOSITORY   TAG       IMAGE ID       CREATED         SIZE
echo_image   latest    208caf6f9272   6 minutes ago   1.09GB
[ec2-user@ip-172-31-47-62 ScratchServer]$
```

```
[ec2-user@ip-172-31-47-62 ScratchServer]$ sudo docker start echo_container
echo_container
[ec2-user@ip-172-31-47-62 ScratchServer]$ sudo docker ps
CONTAINER ID   IMAGE        COMMAND                  CREATED         STATUS         PORTS                                   NAMES
ccb1102e8558   echo_image   "/__cacert_entrypoin…"   8 minutes ago   Up 5 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   echo_container
[ec2-user@ip-172-31-47-62 ScratchServer]$
```

Step Fourteen: View the Results!

Result (a): Webpage



Result (b): telnet

Administrator: Command Prompt

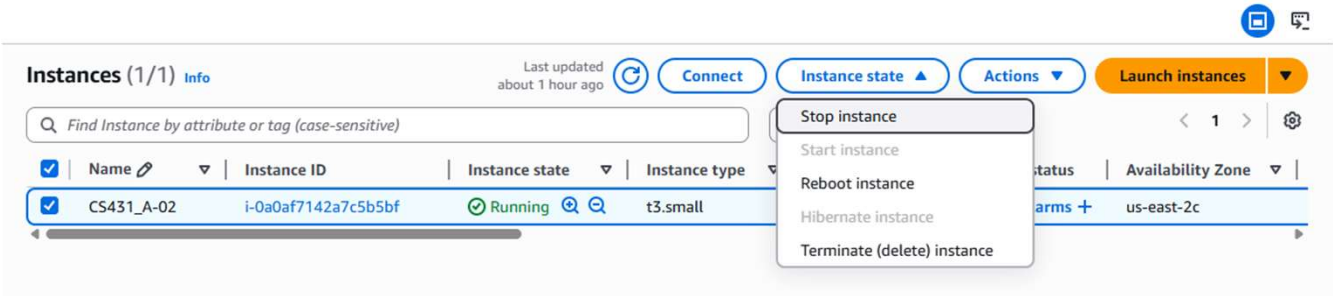
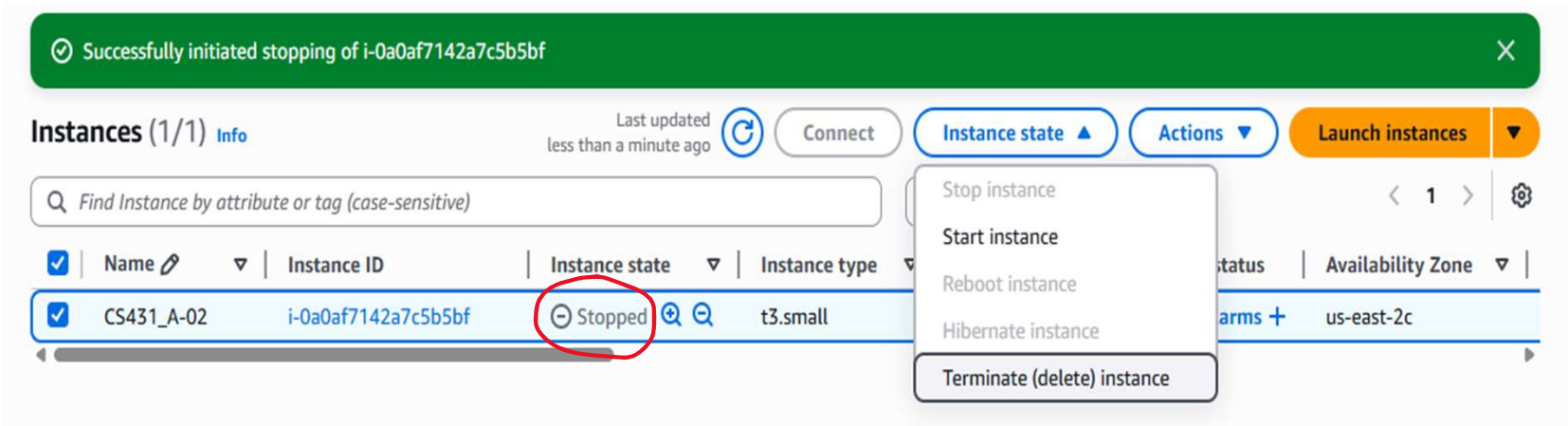
```
C:\Windows\System32>telnet 18.118.171.218 80
```

Telnet 18.118.171.218

```
Hello!
Helloooooooo
Helloooooooo
Hellooooooooooooooo
Hellooooooooooooooo
Heloooooooooooooooooooooooooo
Heloooooooooooooooooooooooooo
ajopsjdifavskvodmjao[sfej0mjgviognhagobvihkdmfvja[pjae[ifjaef
ajopsjdifavskvodmjao[sfej0mjgviognhagobvihkdmfvja[pjae[ifjaef
```

File Edit Selection View Go Run Terminal Help

PROBLEMS DEBUG CONSOLE TERMINAL PORTS OUTPUT

```
[ec2-user@ip-172-31-47-62 ScratchServer]$ sudo docker start echo_container
echo_container
[ec2-user@ip-172-31-47-62 ScratchServer]$ sudo docker ps
CONTAINER ID   IMAGE        COMMAND                  CREATED        STATUS         PORTS                                   NAMES
ccb1102e8558   echo_image   "/_cacert_entrypoin…"    8 minutes ago  Up 5 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp       echo_container
[ec2-user@ip-172-31-47-62 ScratchServer]$
```

Terminate Instance…